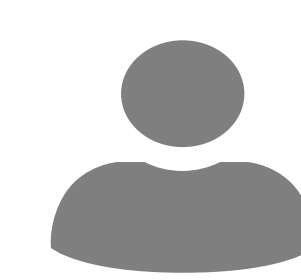


Database cleaning need not be a needle in a haystack

A simple **robust statistics** approach turns thousands of observations into a manageable, reproducible and auditable review process

[NOT] Going MAD Before Hardlock



Marianna Nodale
Simon Bond



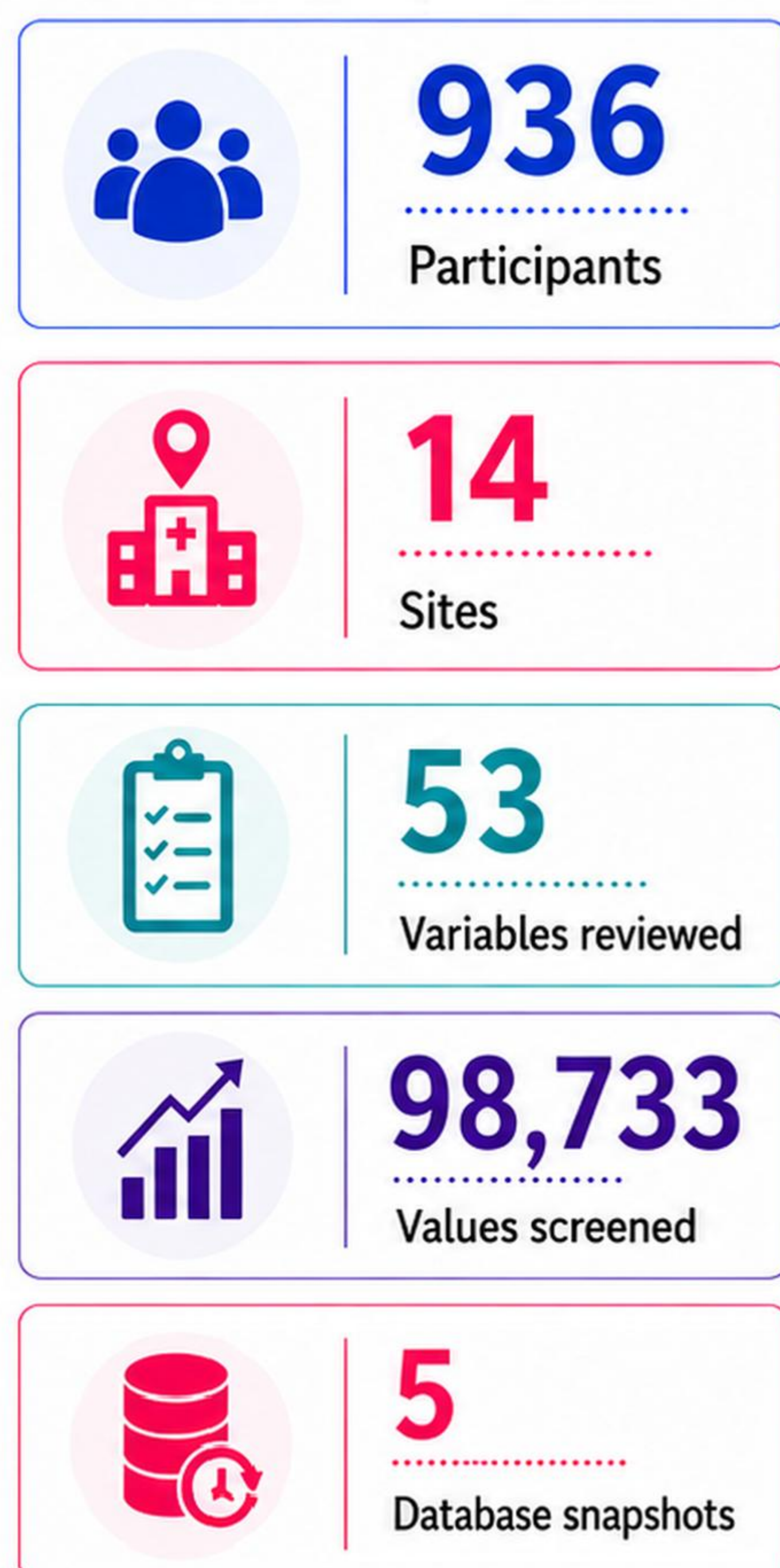
Cambridge Clinical Trials Unit

BACKGROUND

Large multicentre trials often contain heterogeneous data with limited built-in range and logic validation. Database cleaning before hardlock can therefore become a time-consuming "needle in a haystack" exercise, particularly when resources are limited and rapid database lock is required. We developed a simple **robust-statistics workflow** to prioritise observations for review, making database cleaning more focused, reproducible and auditable.

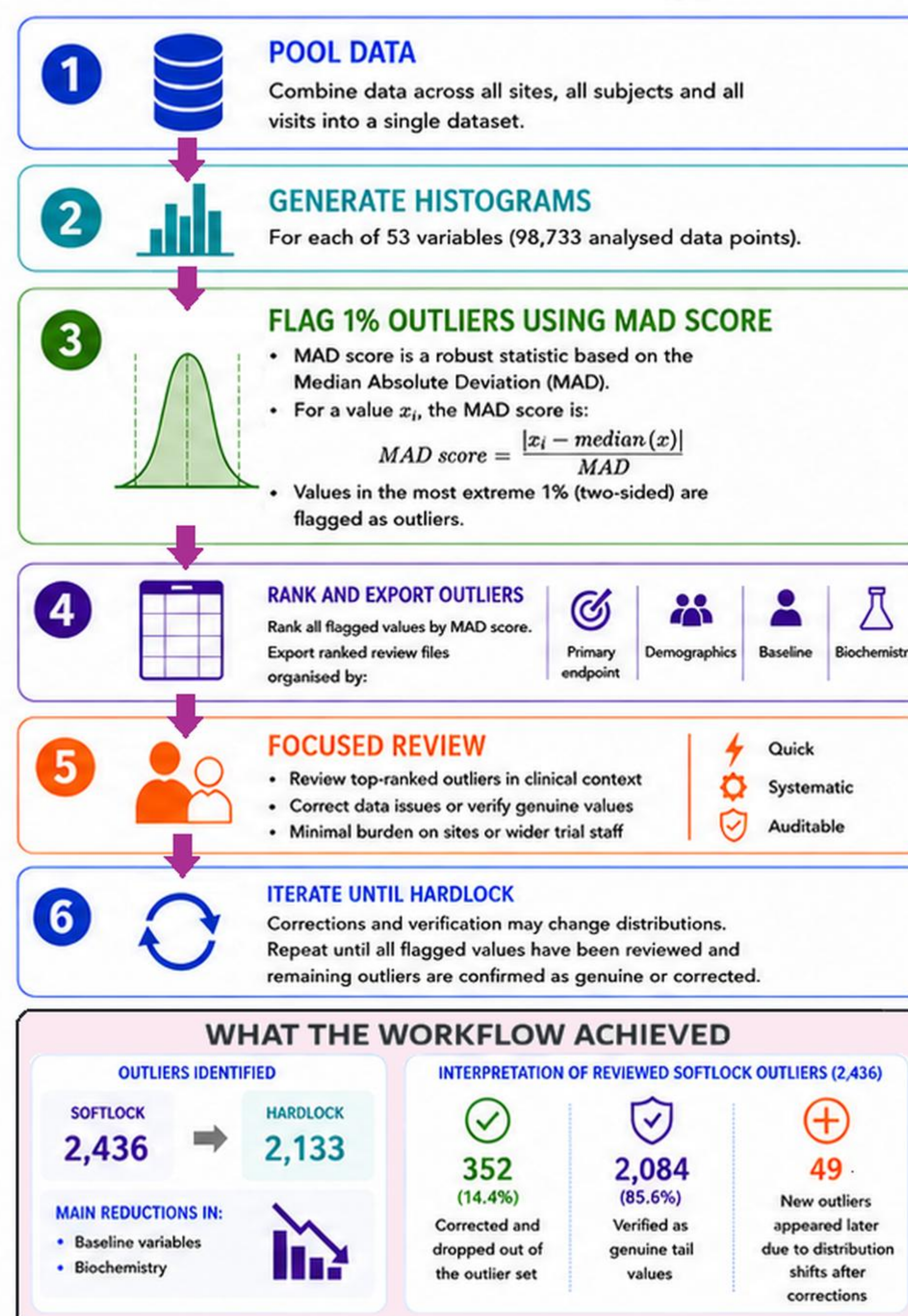
THE TRIAL

Trial at a glance



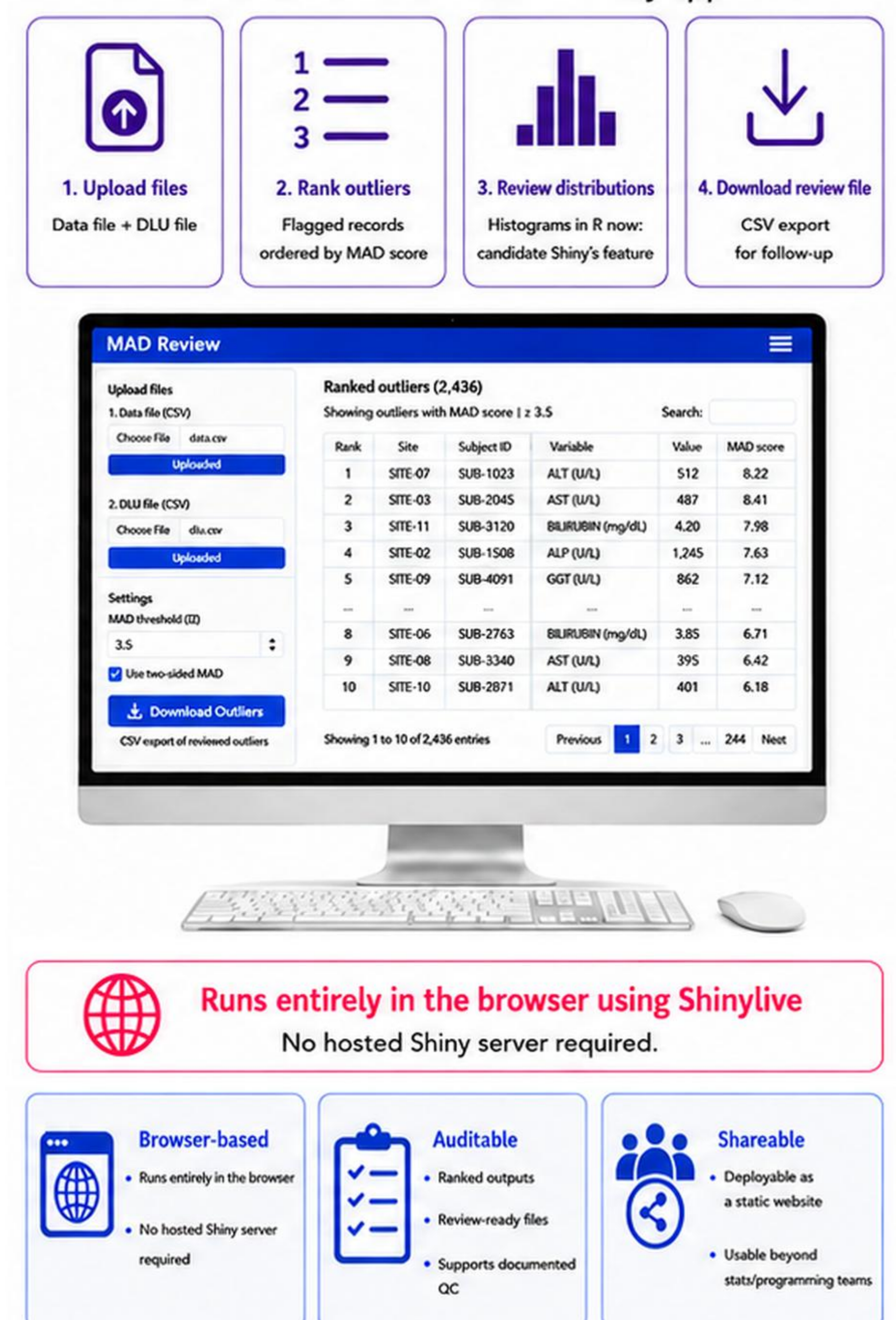
THE WORKFLOW

MAD-based outlier workflow



THE TOOL

From R workflow to reusable Shiny app



WHAT WE LEARNT

- Flagged does not mean wrong.** Most extreme values were genuine and needed confirmation, not correction.
- Simple methods were enough to focus the work.** The gain was operational rather than methodological sophistication.
- Context still matters.** Statistical screening helped identify records for review but did not replace clinical or data-management judgement.
- The workflow was easy to repeat and scale.** A wash-rinse-repeat approach made a large QC task manageable for one data manager and one statistician.

FUTURE DEVELOPMENTS

Extending the tool



➔ Platform for broader **centralised monitoring tools**